Material-technology Modelling the Development of Structures in Wires during Controlled Cooling on Industrial Cooling Conveyors

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Abstract

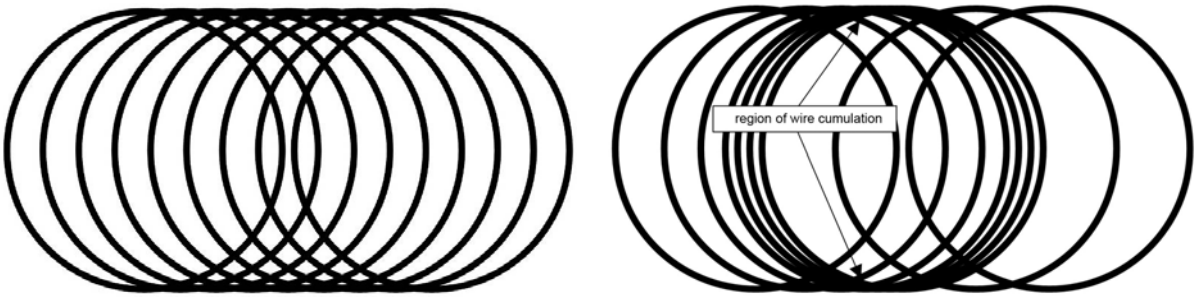
The majority of continuous technological processes for thermomechanical treatments result in a certain amount of variation of mechanical values and structural parameters. Analysis of the causes is very complicated because the influence of materials and technology must be taken into account. One way of identifying the causes is by material-technology modelling. It is shown here that this tool can be used effectively to analyse and quantify the effects of uneven cooling of wire on a real cooling conveyor.

Key words: material-technology model, thermomechanical, simulation, wire, rolling, thermomechanical treatment

1. Material-technology modelling of the development of structures during controlled cooling

A host of technological parameters play an important role during complete material-technology modelling of complex production processes. Some of the most important parameters are not only intensity of deformation and strain rates, but also the thermal profile of the whole process. The aim of this research was to describe the development of structures by material-technology modelling for a range of variants and specific cooling problems after deformation processes which correspond to the technology of wire rolling.
The real process is characterized by the changeable rate of cooling of the wire. This is caused by the uneven intensity of cooling at individual sections of the cooling conveyor. After rolling, the wire is as a rule cooled in a coiled shape in the form of an inclined screw. This causes an uneven energy balance in the cooling process. The loop passes through a coiling head which, at high speeds may occasionally not lay the loops unevenly. (Fig. 1). Movement of the loop to the start of the conveyor generally leads to further uneven laying. This results in loops lying at the edge of the cooling conveyor cooling more slowly than wires lying in the centre of one another in two or even more layers. In contrast, cooling in the centre of the loop is faster because wires lie separately and spaced out. This arrangement, when more loops are piled together, can lead to unfavourable cooling conditions, which can result in localized non-uniform properties along the length of the wire.
This phenomenon is generally known, and occurs to varying extents in practice depending on the type of steel used. In certain circumstances problems with quality can arise which only become apparent during later treatments. In order to describe and minimize this phenomenon, material-technology modelling with a thermomechanical simulator was used. The experiment

was carried out on low alloyed carbon steels alloyed with manganese and boron, which are intended mainly for further cold forming.



optimum distribution of loops on the cooling conveyor

unfavourable, randomly uneven distribution of loops with multiple overlapping

Fig. 1: Distribution of coiled wire on cooling conveyor

1.1. Experimental material

Material-technology modelling was carried out on 19MnB4 and 23MnB4 steels (Tab. 1) , modern low alloyed steels alloyed with only manganese and a small quantity of boron.

Element [%]	19MnB4	23MnB4
C	0.17 – 0.24	0.2 – 0.25
Mn	0.8 – 1.15	0.9 -1.2
Si	0.1	0.3
B	0.001 – 0.005	0.0008 – 0.005

Tab. 1: Guidance chemical composition as set by standards

The default state of both modelled materials has ferrite-pearlite structure with a significant, for rolling, characteristic texture. (Fig. 2., Fig. 3). Samples were taken directly from the actual production process. A specimen for experiment with diameter 23mm with free air cooling was taken directly from the rolling mill.

Material	Ferrite grain size [μm]		Percent ferrite [%]		Hardness			
					HV10		HV30	
	longit. section	transv. section	longit. section	transv. section	longit. section	transv. section	longit. section	transv. section
19MnB4	8.4 ± 5.8	8.7± 6.1	74	77	141	142	138	142
23MnB4	8.8 ± 5.5	8.8± 5.8	75	73	143	150	144	150

Tab. 2: Ferrite grain size, percent of ferrite and hardness in basic state

The ferrite grain size in longitudinal and transverse sections was measured, the percent of ferrite and the hardness HV10 and HV30 was recorded. The ferrite grain size in both sections

varied between 8 and 9 μm (Tab. 2). No marked variation of the percent of ferrite was observed between the longitudinal and transverse sections.
The ultimate strength in 19MnB4 steel in its basic state was 543 MPa, and for 23MnB4 524 MPa (Tab. 3).

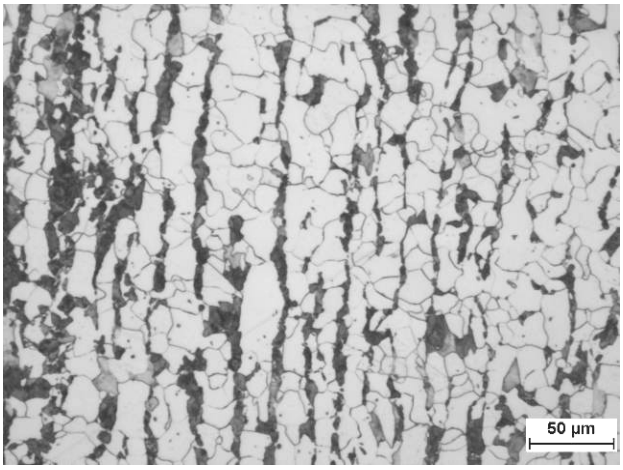


Fig. 2: 19MnB4, longitudinal section, Nital

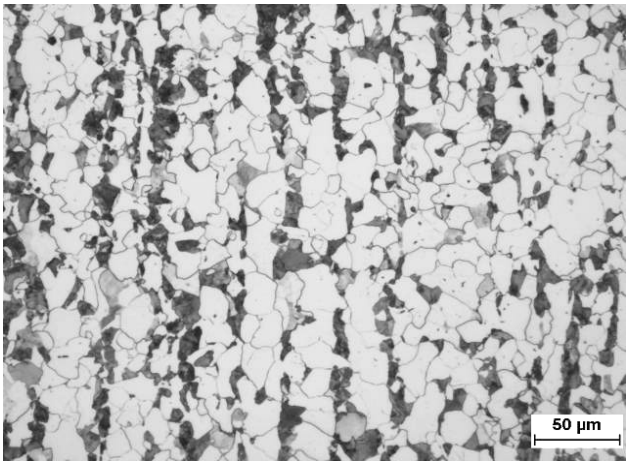


Fig. 3: 23MnB4, longitudinal section, Nital

Material	R_m [MPa]		A_{5mm} [%]		KCV [J.cm^{-2}]	
19MnB4	541	543	48	50	83.6	84
	544		51			
23MnB4	518	524	50	49	83.7	81
	530		48		78.1	

Tab. 3:Mechanical properties of experimental material

1.2. Material-technology model of the production process

The material-technology model of the actual technological process was based on data obtained from measurements on the rolling mill (Fig. 4). Part of the data was obtained by calculation. Treatments were modelled on a thermomechanical simulator. Thermal and deformation changes in the regime were, according to their time dependencies, programmed according to their tabulated descriptions. Specimens were heated by direct resistance heating. Temperature was measured using thermocells welded to the surface of the specimen body.

1.3. Using the material-technology model to specify a problem

The original material-technology model of the technological process was created for mean values of cooling rates. In order to obtain deviations in the structure under the influence of uneven cooling rates at boundaries and in the centre of loops, the relevant data must be obtained. This was done by standard means, measured using a pyrometer with automatic emission correction and laser direction indicator. The cooling rate was modified according to the temperature-time relations for temperatures between 823°C to 390°C in the material-technology model (Fig. 5, Fig. 6). At the same time, three models of the cooling rates were compared which corresponded to the boundary, centre of the loops and mean cooling rate. The hardness, ferrite grain size and mechanical properties of the experimentally obtained structures were then compared. (Fig. 7, Tab. 4, Tab. 5).

Because the wire is coiled randomly it is not possible to take samples from exactly the same place each time. It is only possible to compare states which arose in similar, comparable situations. This could affect the absolute values of the compared parameters. However, it is important that the model reacts with enough sensitivity to the problem. In this case it was found to be sensitive enough to show the deviations in much greater detail than required by the standards.

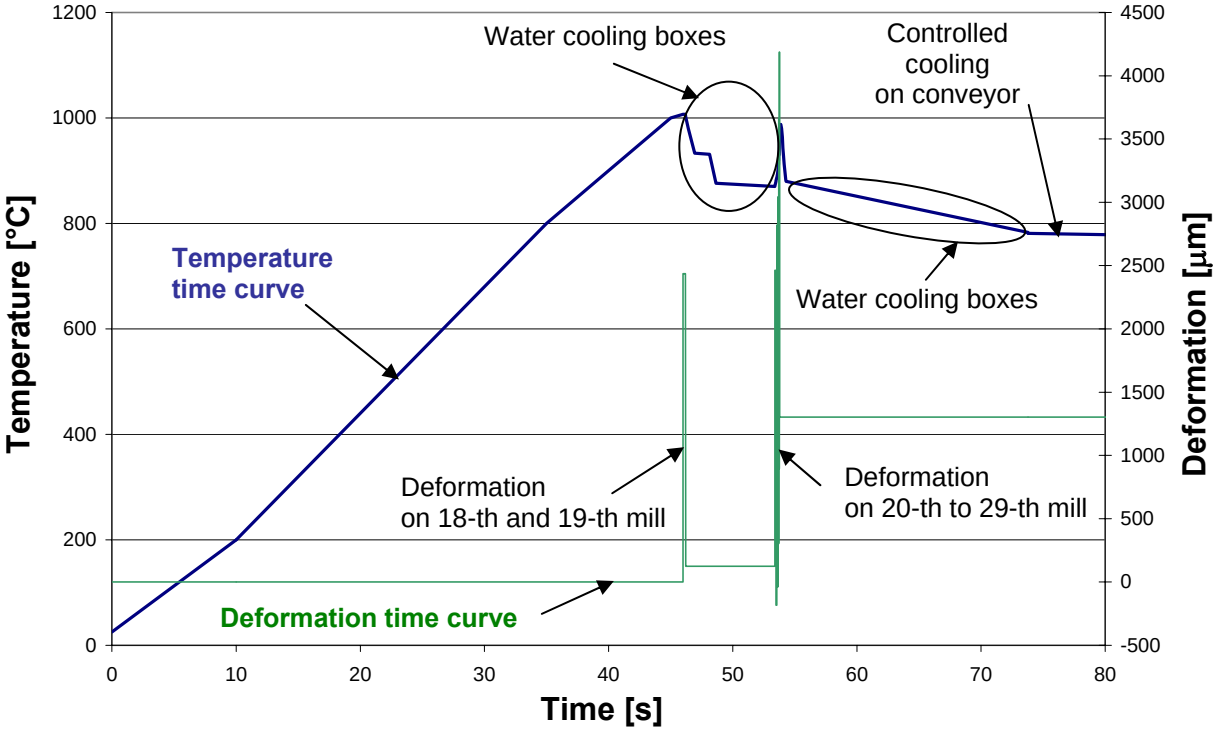


Fig. 4: Time - temperature and deformation diagram from the simulator

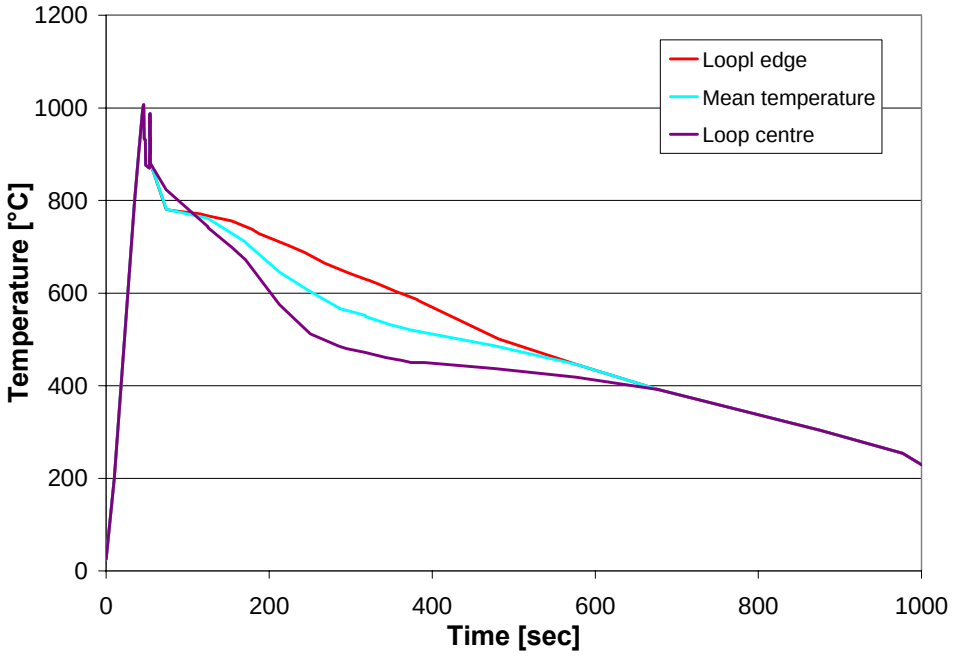


Fig. 5: Differing final cooling curves on the cooling conveyor for models of the loop centre, loop edge and mean cooling temperature

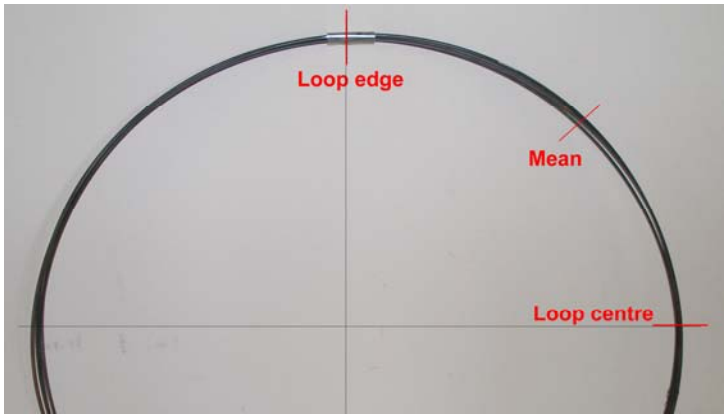


Fig. 6: Sample sites on the loop

	Edge			Mean			Centre		
	Wire	Simu-lation	Deviation [%]	Wire	Simu-lation	Deviation [%]	Wire	Simu-lation	Deviation [%]
HV10	141	141	0	146	149	2	152	157	3
HV30	145	140	-4	149	145	-3	150	156	4
R_m [MPa]	495	501	1	511	515	1	519	524	1
A_{5mm} [MPa]	53	46	-15	58	54	-7	45	49	8
KCV [Jcm^{-2}]	90	87	-3	88	92	4	87	92	5

Tab. 4: Comparison of properties of samples taken from various locations in the loop and under various simulated cooling conditions for 19MnB4

	Edge			Mean			Centre		
	Wire	Simu-lation	Deviation [%]	Wire	Simu-lation	Deviation [%]	Wire	Simu-lation	Deviation [%]
HV10	145	149	3	148	152	3	152	163	7
HV30	146	144	-1	148	152	3	152	161	6
R_m [MPa]	501	503	0	506	509	1	520	524	1
A_{5mm} [MPa]	50	47	-6	48	53	9	51	47	-9
KCV [Jcm^{-2}]	80	86	7	90	92	2	78	89	12

Tab. 5: Comparison of properties of samples taken from various locations in the loop and under various simulated cooling conditions for 23MnB4

As far as material and microstructural properties are concerned, in the model of the loop edge a ferrite-pearlite structure originated with somewhat coarser ferrite grains about 9 μm (Fig. 8, Fig. 9, Tab. 6). These were the coarsest structures observed in all the variants under investigation.
The structures with ferrite grains about 7 um were achieved in the cooling model for mean temperature for both materials (Fig. 10, Fig. 11).
In the cooling model for the loop centre, ferrite grains of 8.5 μm were observed in 19MnB (Fig. 12, Tab. 6) and for 23MnB (Fig. 13, Tab. 6) the grain size was 7.1 μm. Accelerated cooling and the associated finer grain size caused an increase of hardness of 11% when compared to the loop edge.

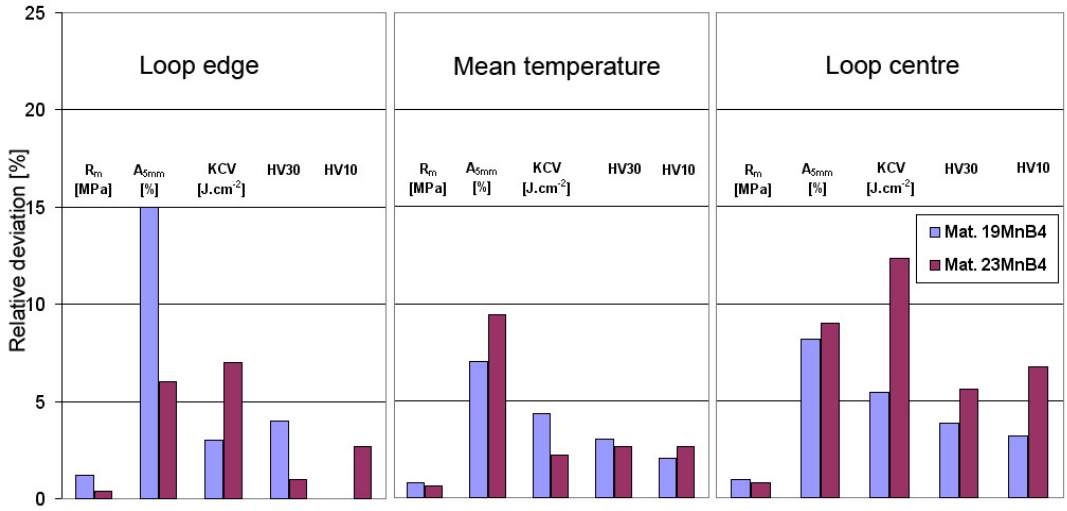


Fig. 7: Relative deviations of observed parameters

model	Material	Ferrite grain size [μm]	Percent ferrite [%]
Loop edge	19MnB4	9 ± 5.9	73
	23MnB4	9.1 ± 6	75
Mean temperature	19MnB4	7.3 ± 4.9	72
	23MnB4	7.1 ± 4.8	71
Loop centre	19MnB4	8.5 ± 5.7	68
	23MnB4	7.1 ± 4.6	65

Tab. 6: Ferrite grain size and percent ferrite for models of loop edge and centre and mean temperature model

Material	Ferrite grain size [μm]		Percent ferrite [%]	
	Longit. section	Transv. section	Longit. section	Transv.section
19MnB4	6.5 ± 5.1	6.5 ± 3.7	77	75
23MnB4	6.2 ± 4	7.1 ± 4.6	78	78

Tab. 7: Ferrite grain size and percent ferrite after completion of technological treatment

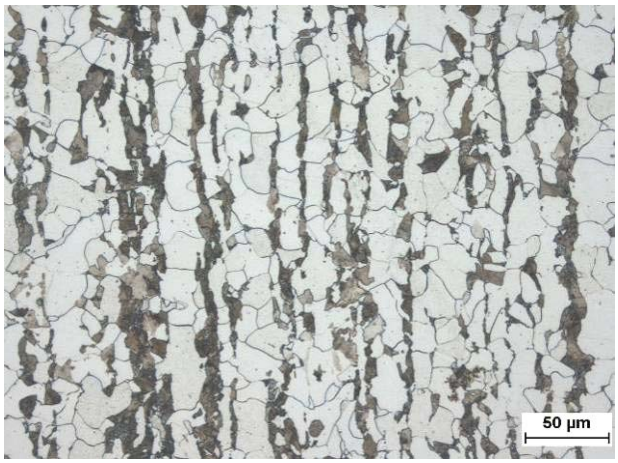


Fig. 8: Model of cooling at loop edge, 19MnB4, longitudinal section, Nital

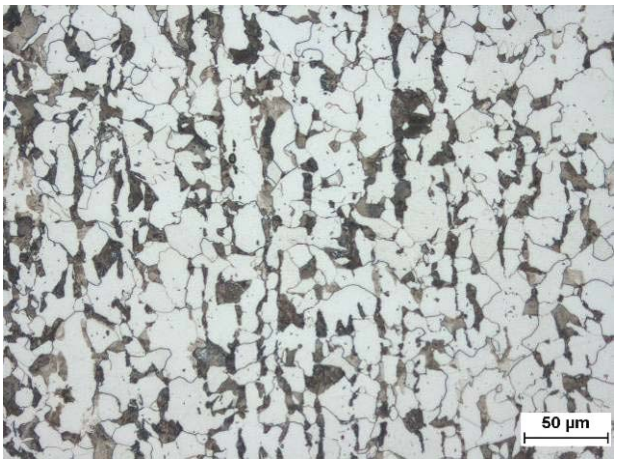


Fig. 9: Model of cooling at loop edge, 23MnB4, longitudinal section, Nital

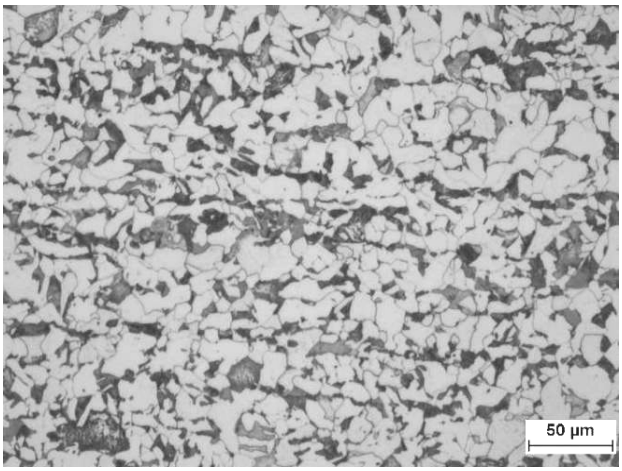


Fig. 10: Model of cooling at mean temperature, 19MnB4, longitudinal section, Nital

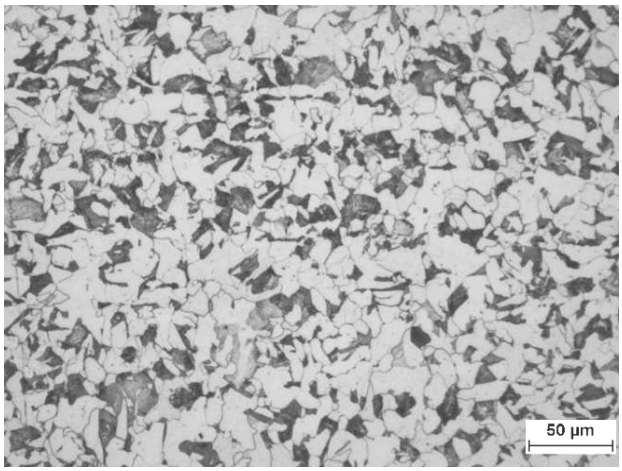


Fig. 11: Model of cooling at mean temperature 23MnB4, longitudinal section, Nital

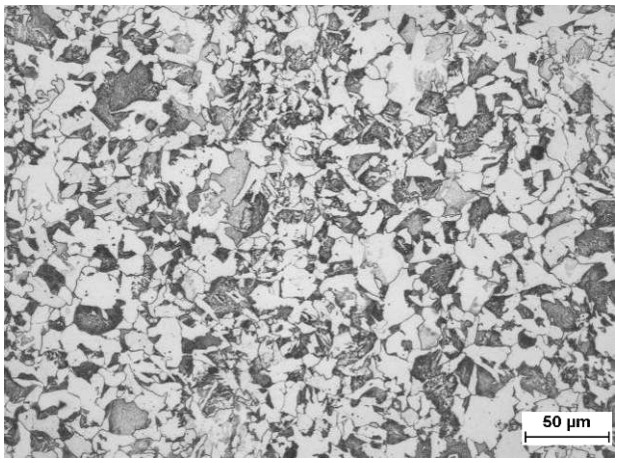


Fig. 12: Model of cooling at loop centre, 19MnB4, longitudinal section, Nital

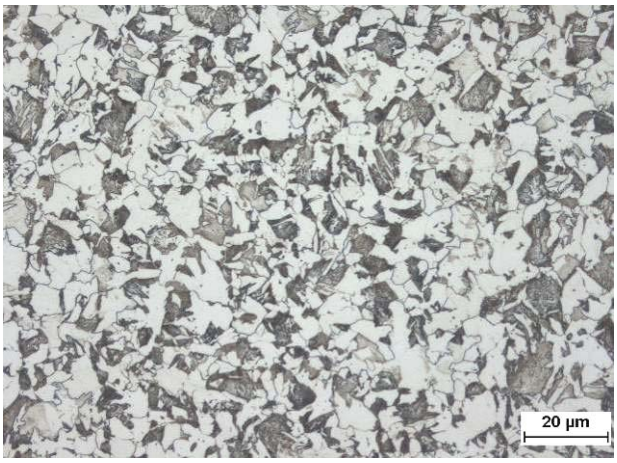


Fig. 13: Model of cooling at loop centre, 23MnB4, longitudinal section, Nital

The mechanical properties of the real and modelled samples were also compared (Tab. 4, Tab. 5). The ultimate strength of the model of cooling at the loop edge was almost identical to the manufactured wire for both types of steel. The largest variations were for ductility values, which were up to 15 %. Deviations for notch toughness were 3 % for 19MnB4 and 7 % for 23MnB4.
There was also good agreement of ultimate strength values between the real product and the model for mean cooling rate. The greatest variations were again found to be for ductility values. The deviation was 7% for 19MnB and 9 % for 23MnB4 (Tab. 4, Tab. 5, Fig. 7).
The same trend was observed for models of the loop centre.

2. Conclusion

Material-technology modelling was used to ascertain the influence of real conditions on the origin of variations in the structure and properties of low-alloyed wire. A model developed for standard thermomechanical treatments was modified to show changes in structure and properties caused by uneven cooling at the edge and centre of wire loops on the cooling conveyor.
It was shown how varying cooling rates on the cooling conveyor influence the structure and properties of 19MnB4 a 23MnB4 steel wire.
The results showed that material-technology modelling can react with sufficient accuracy to variations in parameters which may arise in the real process. The agreement of the model with reality was mostly within 5%. Only in exceptional circumstances did the variation exceed 10%, meaning that none of the observed parameters exceeded a relative deviation of 15%.

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